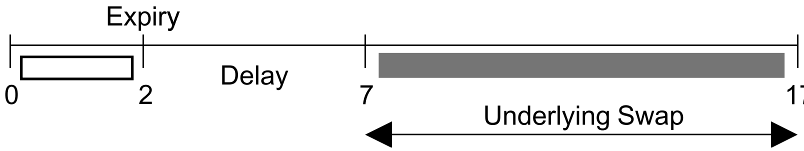

Figure 9.1: A Swaption on a Forward Starting Swap

This swaption expires in two years, and the underlying swap is a 5x15, a 10-year swap, five years forward. That is, the effective date on the underlying swap is five years after the expiry of the option, and the underlying swap has a 10-year maturity. Notice that this is quite a bit different from any other swaption that we have looked at before. In standard swaptions, the effective date of the underlying swap immediately follows the expiry of the option.[7] In this somewhat exotic swaption, there is a 5-year delay from the expiry of the option until the effective date of the swap. Referring now to Table 5.3, which implied swaption vol should be used to price up this swaption?

The answer is that it is not appropriate to use any of these implied vols. The nature of this swaption is such that it does not lend itself well to simply using a single implied vol from Table 5.3. To see this, consider again that the underlying is a 5x15 swap. As we discussed in Section 3.3.4, any forward starting swap can be decomposed into two spot starting swaps. In particular, if we receive on a 5x15 swap, it is equivalent to receiving on a spot starting 15-year swap and paying on a spot starting 5-year swap. So we can start to think of this 2-year option on a 5x15 swap as an option on a portfolio of a 15-year swap and a 5-year swap. Thus, the implied volatility of this swaption depends not only on the implied volatilities of the underlying 15-year and the underlying 5-year swap rates but also on the correlation between them.[8] We will see shortly what this has to do with forward vol, but for now let's just keep it in our back pocket that a swaption on a forward starting swap is a trade that has a correlation component to it.[9]

[7]Recall that in a standard USD swaption the effective date of the underlying swaption is typically two New York and London Business Days after expiry, which is commonly referred to as "$T + 2$."

[8]Movements in the forward starting swap rate depend greatly on movements in each of the two spot starting swaps. Consider again the example of the 5x15 swap rate that can be decomposed into a position in a spot starting 15-year swap and a spot starting 5-year swap. Movements in the 5x15 swap rate are driven by movements in each of the 15-year and 5-year spot starting swap rates. The volatility of the 5x15 swap rate will depend not only on the volatility of the 15-year swap rate and the volatility of the 5-year swap rate, but also on their correlation.

[9]So how does one value this exotic swaption? One way is to ignore the fact that the 5x15 swap rate can be decomposed into two spot starting swaps and just regard the 5x15

9.2.2 Description of Forward Vol

A *forward vol trade* is an agreement to buy or sell some underlying swaption straddle at a given time in the future. The *forward price* is the price at which the underlying swaption straddle will ultimately be bought or sold. It is determined at the time the trade is transacted. The forward price can be either above or below the current spot price of the swaption straddle, depending on the shape of the vol grid. A forward vol trade is a forward contract, and as such no cash flow is exchanged over the life of the contract. At *settlement*, the value of the contract is the difference between the forward price and the then-prevailing spot price for the swaption straddle in the market.

One important feature of a forward vol trade is that the strike on the underlying swaption straddle is not set until the settlement of the contract. At the settlement of the contract, the strike will be set to the then-prevailing at-the-money forward strike on the underlying swaption straddle. It is often thought then that a forward vol contract trade does not have any exposure to movements in rates over the life of the contract, and there are no associated delta nor gamma positions to potentially hedge over the life of the contract. In addition, until the strike is set, there is no concept of theta decay over the life of the contract. Therefore, trading in a forward vol contract is often regarded as taking a pure view on the level of implied vol, as compared to someone who purchases swaption straddles and has associated delta, gamma, and theta positions.[10]

Example

Consider a forward vol trade in which an investor agrees to buy \$100mm 5 into 10 swaption straddles, two years forward. Suppose that the forward price on the straddle is 725 bps, as compared to the current spot price of 5 into 10 straddles, which is 748 bps. Under the terms of the contract, in two years, the investor is committing to buy \$100mm of a then-at-the-money forward 5 into 10 swaption straddle for 725 bps. The strike will be determined in two years by looking at the strike on then-prevailing 5 into 10 swaption straddles struck at-the-money forward that are trading in the market. In other words,

as the underlying itself. This ignores the correlation component of the trade, of course. One would then make a distributional assumption about the underlying and use some model to derive a price. For example, one could assume lognormality, and use a Black's formula–type approach. This approach would not be terribly robust. Moreover, it would be inconsistent with the assumption that each of the underlying 15-year and 5-year spot starting swaps are lognormal.

[10] The statement that a forward vol contract is a pure view on vol (and therefore, in particular, has no exposure to the movement of rates) is not entirely accurate, as we shall discuss later in this chapter.

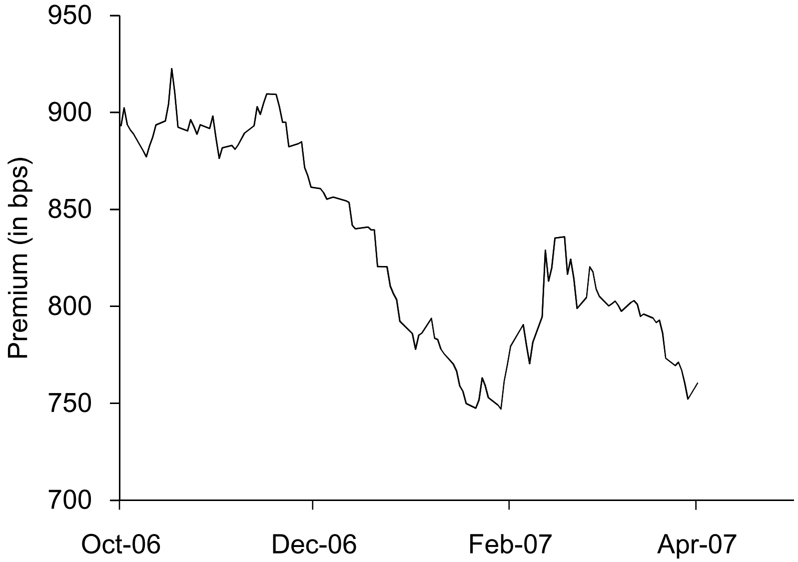

Figure 9.2: Historical 5 into 10 Swaption Straddle Prices

the strike will be set to the then-prevailing 10-year swap rate, five years forward. There will be no cash flow over the first two years of the contract. In two years' time, the payoff to the investor who has bought the forward vol contract is the then-prevailing price on an at-the-money forward 5 into 10 swaption straddle, less the forward price of 725 bps. If, in two years, 5 into 10 swaption straddles struck at-the-money forward are trading above 725 bps, the investor makes money. Otherwise, the investor loses money.[11]

Figure 9.2 shows a history of 5 into 10 swaption straddle prices from October 2006 through April 2007. The graph shows the price, in basis points, of a 5 into 10 swaption straddle that is struck at-the-money forward each day during this period. As shown in this figure, 5 into 10 at-the-money forward swaption straddles did not trade much below 750 bps during this six-month period of time. An investor who could buy 5 into 10 at-the-money forward swaption straddles, two years forward at a price of 725 bps (below anything seen in recent history) may have found this an attractive opportunity.[12] □

[11]Forward vol trades can be either cash settled or physically settled. Depending on opportunities in the market, investors may wish to either buy or sell forward vol.

[12]Actually, we did not see prices for 5 into 10 swaption straddles struck at-the-money forward trade anywhere near the 750 bp area again for several years thereafter. In fact, in the fall of 2011 the price of these 5 into 10 swaption straddles sometimes exceeded 1500 bps, more than double the price observed at times in early 2007. The substantial price

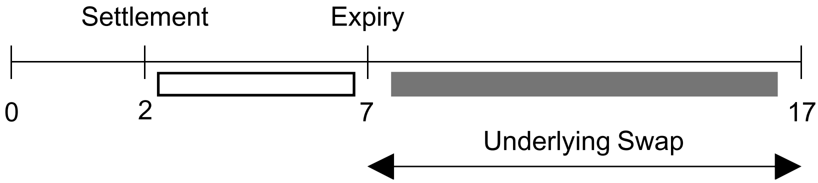

Figure 9.3: A Forward Vol Trade: 5 into 10 Swaption Straddles, Two Years Forward

9.2.3 Heuristic Pricing of Forward Vol Trades

Above all, it should be noted that the pricing and behavior of forward vol trades are very model dependent.[13] Developing a fully rigorous methodology to price forward vol trades is beyond the scope of this discussion. However, in this section we will present a heuristic guide to the pricing of forward vol trades. Let's go back to the example that we considered earlier: a 5 into 10 swaption straddle, two years forward. This forward vol trade is depicted in Figure 9.3.

In this figure, the white box indicates the period of time that the investor is long vol, and the gray box represents the underlying swap. Note that the investor who buys 5 into 10 straddles, two years forward does not have any significant exposure to realized volatility (i.e., movements in rates) over the first two years during the life of the contract since the strike on the underlying straddle has not yet been set.[14] However, during the two-year life of the contract, the trade has exposure to changes in implied vol. For example, suppose an investor enters into a forward vol contract in which he

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increase of these swaption straddles was due both to a pronounced increase in (normal) implied vol during this time and to a significant drop in swap rates. Under the normal model, the price of a swaption straddle struck at-the-money forward is

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A$$

where T is the time to expiry of the swaption, σ_N is normal implied vol, and A is the present value of a 1 bp annuity over the life of the underlying swap. As swap rates fall the value of the annuity A increases. As is evident from this equation, both an increase in normal implied vol and a decrease in rates serve to increase the value of an at-the-money forward swaption straddle under the normal model.

[13]The characterizations of how forward vol trades behave that we note throughout this chapter are in general agreement with most pricing models used throughout the industry. That being said, certain models may provide results that do not concur with all of the observations made in this chapter.

[14]As we will discuss, forward vol trades based on swaption straddle premia have a rate bias, and thus one could argue that such forward vol trades do have exposure to realized vol.

agrees to buy vol forward. Suppose shortly thereafter all normal implied vols in the vol grid rise by an equal amount. In general then the contract will be in-the-money to the investor on a mark-to-market basis.[15]

This figure is somewhat reminiscent of the work we did in Chapter 3 on forward starting swaps by virtue of the fact that there is a period of exposure in the future following a period of delay during which there is no exposure. Recall our basic methodology for computing a forward starting swap rate. For example, in order to compute the 3-year swap rate, two years forward (a 2x5), we discussed that we could receive in a spot starting swap for five years and simultaneously pay for two years. The net effect of this was (with some manipulation) to create no cash flows over the first two years and equal cash flows over the subsequent three years on the fixed rate side, which led us to the 2x5 swap rate we desired.

The same logic can be applied to replicating a forward vol trade. In order to recreate the 5 into 10 swaption straddle, two years forward, consider buying 7 into 10 at-the-money forward swaption straddles today and simultaneously selling 2 into 10 at-the-money forward swaption straddles, as shown in Figure 9.4. In this figure, the white boxes represent periods of time in which we are long vol, and the black box represents periods of time during which we are short vol. By buying 7 into 10 swaptions and selling 2 into 10 swaptions (in the right proportions), our vol exposure over the first two years "cancels" out, and we are left with forward vol exposure. The gray box in the top line of the figure represents the underlying forward rate on which the forward vol trade is written. This is the same forward rate on which 7 into 10 swaption straddles are written, corresponding with the gray box in the middle part of the figure. In the bottom part of this figure, corresponding to the sale of 2 into 10 swaption straddles, the gray box represents the forward rate on which a 2 into 10 swaption straddle is written. But this is not the forward rate that we wish to recreate. Therefore, this strategy of buying 7 into 10 straddles and selling 2 into 10 straddles does not perfectly replicate the forward vol trade, as there is a mismatch in the underlying forward rate that is in question.

We can, however, consider modifying this replication somewhat to address this mismatch. Instead of selling 2 into 10 swaption straddles as in Figure 9.4, consider instead the replication shown in Figure 9.5. In this replication, the short 2 into 10 swaption straddles have been replaced by 2 into 5x15 swaption straddles. A 2 into 5x15 swaption straddle is a com-

[15]It may be useful to think of the following analogy. Consider our stylized example and the swap curve in Table 3.5. We have previously shown that the 1x4 swap rate is 5.644%. If after paying fixed in a 1x4 at this rate the entire swap curve should increase by a parallel shift, the swap will be in-the-money to the investor.

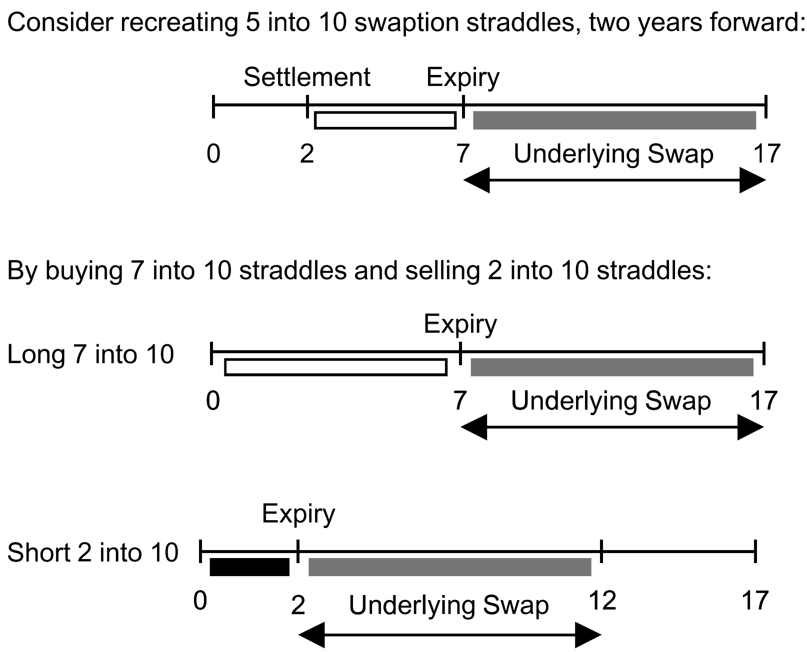

Figure 9.4: Replicating the Forward Vol Trade—Part I

bination of a payer and receiver swaption on a forward starting swap, as we discussed in Section 9.2.1. By making this modification in the replication, both the periods of vol exposure and the underlying forward rate on which the contract is written match up. In Section 9.2.1, we discussed that a swaption on a forward starting swap had a correlation component to it. Thus, as an immediate consequence we see that a forward vol trade must also have a correlation component in the pricing of the contract (provided that we do not regard the underlying forward starting swap in the bottom part of Figure 9.5 as a primitive, of course).

Does the replicating strategy in Figure 9.5 perfectly recreate the forward vol trade? Not exactly. The 7 into 10 straddles and the 2 into 5x15 straddles will have strikes that are set today, whereas the forward vol trade does not have strikes that are set until settlement in two years. This implies that the replicating strategy will have associated delta, gamma, theta, and so on, over the first two years, whereas the forward vol trade does not. Thus, this heuristic pricing model does not capture all aspects of the forward vol trade. To put it another way, the forward vol trade will have its strike (which is determined on the settlement date) set at-the-money forward, whereas the

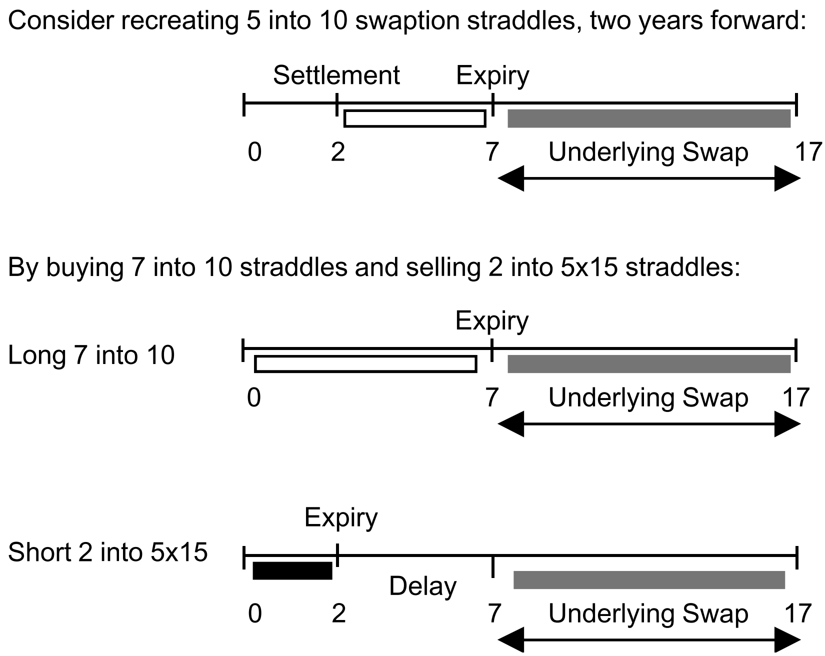

Figure 9.5: Replicating the Forward Vol Trade—Part II

strikes in the replicating strategy are set today and thus will potentially be away-from-the-money at settlement. Thus, the replicating strategy may have skew considerations, whereas the forward vol trade will not.[16]

9.2.4 Will the Forward Price Be Higher or Lower Than the Spot Price?

The answer to this question depends on the shape of the vol grid, not to mention the pricing model employed. Investors who are interested in buying forward vol will often look for trades in which the forward price is lower than today's spot price. Conversely, investors interested in selling forward vol are

[16]In two years at settlement, the strike will be set on the underlying swaption in the forward vol trade to the then-prevailing at-the-money forward strike. Thus, there will be no skew associated with the forward vol trade at settlement. However, in the replicating strategy, the strike on the 7 into 10 swaption straddle will be set today. Thus, in two years when valuing what is then a 5 into 10 swaption straddle, the strike may be significantly away-from-the-money, which implies that the value of the option will depend on skew.

		Underlying Swap (Tail)									
		1	2	3	4	5	7	10	15	20	30
	1m	68.6	74.9	73.4	72.4	71.4	68.6	63.0	60.2	59.0	57.8
	3m	63.3	73.2	71.2	71.1	70.5	68.1	64.1	62.3	60.3	59.2
E	6m	71.9	75.5	74.0	73.0	72.4	70.2	67.0	64.7	63.1	61.8
x	1y	78.1	77.1	76.1	75.0	73.9	72.0	69.1	66.5	65.1	63.6
p	2y	80.3	80.0	79.0	77.7	76.8	74.8	**71.7**	68.8	67.3	65.9
i	3y	81.3	80.0	79.3	78.5	77.8	75.8	72.8	69.5	68.0	66.1
r	4y	80.6	80.4	79.7	78.9	78.0	76.1	73.3	69.7	67.9	65.9
y	5y	81.6	80.2	79.5	78.7	78.0	76.0	73.2	69.4	67.7	65.5
	7y	78.7	77.5	77.1	76.4	75.9	73.9	**71.0**	66.9	64.3	62.1
	10y	74.3	73.3	73.0	72.7	72.3	70.2	67.3	62.9	59.8	57.7

Table 9.1: Normal Implied Vol Swaption Matrix

often interested in forward vol trades in which the forward price is above today's spot price. Let's look at a normal implied vol swaption matrix, such as in Table 9.1. Refer now to Figure 9.4 in which we suggested as our first (and far from exact) stab at recreating a 5 into 10 swaption straddle, two years forward that we might buy some 7 into 10 straddles and sell some 2 into 10 straddles. Despite the inaccuracies in this approach, it turns out from a practical point of view that if we compare 7 into 10 normal implied vol with 2 into 10 normal implied vol, this will usually give us a good indication as to whether most pricing models used in industry will indicate that forward vol is trading at a discount or premium to spot. For example, as shown in Table 9.1, 7 into 10 is cheaper (in normalized vol terms) than 2 into 10, thereby suggesting that the forward price will be less than today's spot price. As we discussed in Section 9.2.2, 5 into 10 swaption straddles had a spot price of 748 bps, whereas the forward price was only 725 bps.

9.2.5 Are Forward Vol Trades Truly a Pure View on Vol?

Not exactly. Recall that under the normal model, the value of an at-the-money forward swaption straddle is

$$\sqrt{T} \times \sqrt{\frac{2}{\pi}} \times \sigma_N \times A, \tag{9.1}$$

where T is the time to expiry of the swaption, σ_N is normal implied vol, and A is the present value of a 1 bp annuity over the life of the underlying swap. Under the normal model, as the market sells off or rallies, even if σ_N remains constant (as the normal model assumes to be the case), the value of the swaption will still change since the value of the annuity, A, depends on the level of rates. Therefore, as the market rallies and the value of the annuity A goes up, so will the value of the swaption, and vice versa. Thus, swaption straddle prices have an inherent rate bias. The implication of this is that forward vol trades do not represent a pure view on vol. For example, the buyer of forward vol will benefit, all else equal, as the market rallies as that effect in and of itself will raise the price of swaption straddles.

In order to avoid this rate bias, some investors prefer to quote their forward vol trades in terms of normal implied vol, σ_N, instead of swaption straddle prices. From Equation 9.1, we have

$$\sigma_N = \frac{\text{Straddle Price}}{\sqrt{T} \times A} \times \sqrt{\frac{\pi}{2}}. \tag{9.2}$$

In this parameterization, a forward vol trade is a forward contract on the normal volatility of some underlying swaption straddle. Instead of quoting a forward price, we quote a forward normal vol on an underlying swaption straddle and define a set dollar amount for every basis point that normal implied vol is above or below the forward vol. For example, suppose an investor buys forward vol on 5 into 10 swaption straddles, two years forward. The investor buys vol forward on a normal vol basis, instead of on a price basis, and locks in a level of 70.7 bps. Assume that the investor risks \$100,000 per bp of normal implied vol. The payoff to the investor in two years is \$100,000 × (NIV − 70.7), where NIV is the normal implied vol of the underlying 5 into 10 swaption straddle in two years.[17]

9.2.6 Bermudan Cancelable Swaps Revisited

In Section 6.2.3 we considered a 3nc1 Bermudan cancelable swap in which we pay fixed for three years and have the right to cancel in one year and again in two years. We showed that our decision to cancel the trade in a year depended (in part) on where 1 into 1 swaption vol was prevailing in a year. We considered a case in which we would cancel the Bermudan in a year if

[17] The normal implied vol, NIV, of the swaption straddle is determined at the time of settlement using Equation 9.2 and the then-prevailing spot price of the swaption straddle.

then-prevailing 1 into 1 swaption vol were low and another case in which we would not cancel the Bermudan in a year if then-prevailing 1 into 1 swaption vol were high. When 1 into 1 swaption vol prevailing in a year was low, we weren't leaving much on the table by canceling early and forgoing the second option. With a higher level of then-prevailing 1 into 1 swaption vol, it was advantageous not to cancel so as to recognize the value of that option.

We concluded that the value of the 3nc1 Bermudan cancelable depends (in part) on where we think 1 into 1 swaption vol would be in a year. The higher that we think that 1 into 1 swaption vol will be in a year, the more we think that we will likely keep the trade on for an additional year and not cancel it at the first cancelation date. By not canceling in a year, we will be able to recognize the value of the remaining embedded option. Therefore, the higher we think that 1 into 1 swaption vol will be in a year, the greater value there is to the 3nc1 Bermudan cancelable.

This concept of where we "think" that 1 into 1 swaption vol will be prevailing in a year is an example of forward vol. There is no reason to think that 1 into 1 swaption vol prevailing in a year will be the same as today's 1 into 1 swaption vol. The level of 1 into 1 swaption vol prevailing in a year can be either less than, greater than, or equal to today's 1 into 1 swaption vol. Just as forward swap rates are sometimes interpreted as the market's expectation of where swap rates will be in the future, forward vol is sometimes interpreted as the market's expectation of where swaption vol will be in the future. Thus, the valuation of Bermudan cancelables is inexorably linked to a consideration of forward vol. The higher 1 into 1 swaption vol, one year forward is, the greater the value of the 3nc1 Bermudan cancelable.

The relatively recent advent of forward vol as a stand-alone trade has its roots in the longer-established market for Bermudans. Traders who dealt in Bermudans for years before the rise in popularity of forward vol were well aware of the links between the two. And the issue of correlation risks that are implicit in forward vol has been well understood by many to have a role in the valuation of Bermudans.

9.3 Curve Options

A *curve option* or *CMS spread option* is an option whose payoff depends on the difference between two different CMS rates.[18] For example, a (linear) curve cap on 2s10s has a payoff that depends on

[18] As a reminder, CMS stands for Constant Maturity Swap. A CMS rate of a given maturity is a floating rate of interest. For example, 10-year CMS is the floating rate of interest equal to the 10-year swap rate.